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SUPERSEDING
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November 5, 1957

FEDERAL SPECIFICATION

HAMMERS, PNEUMATIC, PORTABLE (CHIPPING AND SCALING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable pneumatic hammers for use in scaling and chipping, that are most generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, styles, and sizes. Portable pneumatic hammers shall be of the following types, styles, and sizes, as specified (see 6.1):

Type I - Scaling.
  Style A - Single (nozzle type).
  Style B - Triple pistons and chisels integral.
  Style C - Single piston and chisel integral.

Type II - Chipping
  Size 1
  Size 2
  Size 3

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth. Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-P-12829 - Packaging of Pneumatic Hand Tools with Attachments and Accessories.
MIL-S-22698 - Steel Plate: Carbon, Structural, for Ships.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are descriptive and not restrictive and are not intended to preclude the purchase of hammers otherwise conforming to this specification.

3.2 Qualification. Portable pneumatic hammers furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on, or approved for listing on the applicable qualified products list (AQL). (For information concerning qualification on application see 6.2.)

3.3 Material. Material used in the construction of hammers shall be of a good commercial quality suitable for the intended purpose. All materials used shall be free from defects and imperfections.

3.4 Screw threads. All threaded parts including screws, bolts, and nuts shall conform to the applicable requirements of NBS H28.

3.5 Identification of product. Hammers shall be furnished with the manufacturer's standard nameplate. The nameplate shall be marked in a plain and permanent manner so that the part number and source of manufacture can be easily determined.

3.6 Lubrication. Hammers shall be provided with an oil reservoir. The oil reservoir shall be of sufficient capacity so that one filling will supply lubrication to moving parts in the path of the air stream for at least 4 hours of continuous operation. An automatic built-in oiling device shall feed the oil from the reservoir during operation of the hammer.

3.7 Air inlet connection. The air inlet connection shall consist of a well-secured ferrous fitting threaded with internal American Standard taper pipe threads (NPT) or internal Dryseal American Standard taper pipe threads (NPTF) in the size specified herein. The air inlet shall be fitted with an air strainer or screw (number 20 mesh wire size minimum) which shall be effective in retaining solid particles in the compressed air supply. The strainer or screen shall be of such design that it can be easily removed for cleaning.

3.8 General construction. Hammers shall be of rugged construction so as to withstand, without failure, treatment likely to be encountered under general service conditions. Hammers shall be self-contained, readily accessible for adjustment or replacement of parts. Parts which require lubrication shall be properly enclosed so as to prevent entrance of foreign particles and leakage of lubricant. Working parts exposed to wear shall be of sufficient hardness to withstand the service required. Parts shall be properly finished by machining or grinding to dimensions within limits established by good commercial practice. Similar parts shall be interchangeable and of good fit when assembled with relative parts of the same size, type, and make. Hammers shall be reliable and effective when operated at gage pressures between 80 and 100 pounds per square inch (p.s.i.).

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3.9 Type I, scaling hammers.

3.9.1 Style A, single (nozzle type). Style A, scaling hammers shall be suitable for scaling paint and rust, and shall deliver a minimum of 3,000 blows per minute when tested under a 25-pound load. The hammers shall be free from projections that interfere with operation in confined spaces.

3.9.1.1 Handle. The handle shall be of the open or closed style. The handle and cylinder barrel shall be locked together, by means of an adjustable locking device, in such a manner that they will not become loose in service. The air-inlet connection in the end of the handle shall be 1/4-inch nominal pipe size.

3.9.1.2 Throttle. A nonleakable, manual throttle shall be provided for admitting, regulating, and shutting off the supply of compressed air. The throttle lever shall be of the conventional outside trigger style.

3.9.1.3 Barrel and piston. The barrel and piston shall be of hardened steel suitable for the purpose intended. Barrels shall be fully heat-treated to a Rockwell hardness of not less than 50 nor more than 62 on the C scale.

3.9.1.4 Style A hammers shall conform to table I and shall be similar to figure 1.

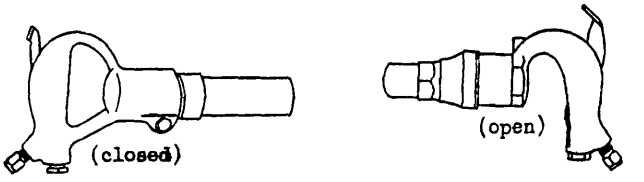


FIGURE 1. Type I, style A hammer, scaling, single (nozzle type).

TABLE 1. Type I, style A, single (nozzle type), scaling hammer.

Weight (max.)	Length overall (approx.)	Diameter of piston	Stroke of piston	Air consumption under load at 90 p.s.i. (max.)
Pounds	Inches	Inch	Inches	Cubic feet
7	9	3/4 to 1	3/4 to 1-3/8	20

3.9.1.5 Chisel bushings. Style A hammers shall be fitted with bushings for receiving chisel shanks, as shown on figure 2.



FIGURE 2. Chisel shank.

(a) Diameter of shank at A: 0.495 ± 0.002-inch.
(b) Inside diameter of bushing 0.500 ± 0.001-inch.

3.9.2 Style B, triple pistons and chisels integral. Style B, triple scaling hammers shall be suitable for scaling paint and rust, and shall deliver a minimum of 12,000 blows per minute (4,000 blows each piston) when tested under a 25-pound load. The triple scaling hammer shall consist essentially of a line throttle, a dead handle, body, and three pistons.

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3.9.2.1 Line throttle handle. The line handle shall consist of a length of pipe having a suitable connection to securely attach the handle to the body of the scaler. The end of the handle opposite the connection shall be fitted with a nonleakable manual throttle for easy control and regulation of the air supply. The throttle shall be either the stopcock style or the lever style and shall withstand 150-pounds air pressure under water without leaking. The line handle shall be fitted with a 3/8-inch nominal pipe size air inlet connection.

3.9.2.2 Dead handle. The dead handle shall be approximately 6 to 8 inches long and of a diameter to furnish a comfortable grip.

3.9.2.3 Body. The body shall be of forged steel suitably heat-treated for the intended purpose and so designed to house and operate three replaceable pistons. The body shall have both a side and end air inlet.

3.9.2.4 Pistons. The pistons shall be steel, hardened to not less than 40 nor more than 62 Rockwell on the C scale, and tipped with "Stellite" or equal, on the striking end, which shall have a flat cross chisel point.

3.9.2.5 Air. The maximum allowable air consumption under full load shall be 25 cubic feet per minute.

3.9.2.6 Type I, style B, hammer shall be similar to figure 3.

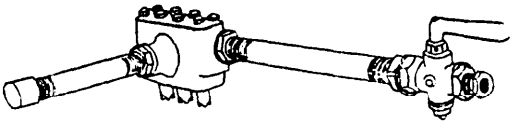


FIGURE 3. Type I, style B, hammer scaling; triple pistons and chisels.

3.9.3 Style C, single piston and chisel integral. Style C, scaling hammers shall conform to the requirements of style B, except that the dead handle shall be furnished at the option of the manufacturer and it shall be a single piston type and deliver approximately 4,000 blows per minute when tested under a 25-pound load. Type I, style C, hammers shall be similar to figure 4.



FIGURE 4. Type I, style C, hammer, scaling; single piston and chisel integral.

3.10 Type II, chipping hammers. Chipping hammers shall be similar to figure 5 and shall be suitable for chipping and calking metals. Except as specified herein, chipping hammers shall conform to type I, style A, scaling hammers, and the requirements of table II.

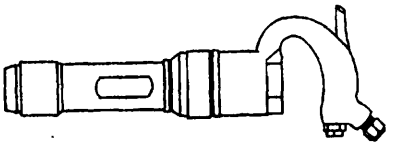


FIGURE 5. Type II, hammer, chipping.

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TABLE II. Type II, chipping hammer.

Size (length of piston stroke inches)	Nature of work	Weight (max.)	Length overall (max.) [1]	Diameter of piston	Steel chipped (min.) [2]	Air consumption under load at 90 p.s.i. (max.)	Blows per minute (min.) when chipping
		Pounds	Inches	Inches	Grams per minute	Cubic feet	
1	Light chipping and calking	12-1/4	13	1-1/6 to 1-1/8	54	30	2700
2	Medium chipping and calking	14-1/2	14-1/2	1-1/16 to 1-1/8	100	34	1950
3	Heavy chipping, calking, and driving rivets up to 1/2-inch diameter	16	15-3/4	1-1/6 to 1-1/8	120	36	1650

[1] Overall length is to be measured exclusive of any bushing in the handle at air inlet.

[2] The chipping test is to be conducted as specified in 4.4.1.

3.10.1 Cylinders. The cylinder barrel shall be fully heat-treated to a Rockwell hardness of not less than 45 nor more than 63 on the C scale.

3.10.2 Chisel bushings. Chipping hammers shall be fitted with bushings for receiving the chisel shanks as shown on figure 6 and stated below.

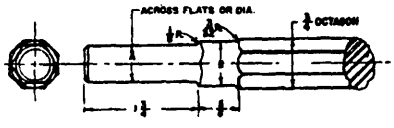


FIGURE 6. Chisel shank.

(a) Nozzle bushing shall be combined round and hexagonal and shall be suitable for use with chisels having either combined round and hexagonal shanks, or two diameter round shanks of the dimensions shown of figure 6.
(b) Dimensions of hexagonal shank, across flats at A: 0.58 ± 0.002-inch.
(c) Diameter of round shank at A shall be either 0.58 ± 0.002 or 0.68 ± 0.002-inch, as specified in the contract or order (see 6.1.)
(d) Dimensions of hexagonal portion of bushing in way of shank at A: 0.592 ± 0.002-inch.
(e) Diameter of shank at B: 0.68 ± 0.001-inch. (This diameter is common to both types of shank.)
(f) Inside diameter of bushing in way of shank at B: 0.6835 ± 0.001-inch.

3.11 Accessories or extra equipment. When accessories or extra equipment are required they shall be as specified (see 6.1).

3.12 Instruction book. One copy of an instruction book consisting of the manufacturer's standard commercial instruction and parts list shall be furnished with each hammer.

3.13 Workmanship. The hammers shall be of the best quality, free from all imperfections which will adversely affect the general appearance, functions, or serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification shall consist of the examination in 4.3.2 and the tests in 4.4 and 4.5.

4.2.1 Sample for qualification. The manufacturer shall submit one hammer for each style (type I), and one hammer for each size (type II), for which qualification approval is desired.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the examination in 4.3.2, the performance tests in 4.5, and the inspection or preparation for delivery in 4.7.

4.3.1 Inspection lot. Hammers of the same type, style, and size presented at one time shall be considered a lot for the purpose of quality conformance inspection.

4.3.2 Sampling for examination. A random sample of hammers shall be selected, in accordance with MIL-STD-105, and visually and dimensionally examined and weighed. The inspection level shall be level II, with an acceptable quality level (AQL) of 2.5 defects per hundred units for visual examination, and level S-4 with an AQL of 1.5 defects per hundred units for dimensional and weight examination.

4.3.3 Sampling for quality conformance tests. A random sample of hammers shall be selected, in accordance with MIL-STD-105, and subjected to the performance tests specified in 4.5 and 4.6 to verify compliance with this specification. The inspection level shall be level S-4 with an AQL of 1.5 percent defective.

4.4 Qualification tests. The qualification tests shall be run as specified below. The test material shall be steel in accordance with MIL-S-22698.

4.4.1 Load tests, type I scaling hammers (styles A, B, and C). Scaling hammers shall be tested to determine the number of blows per minute delivered while under the applicable load specified for the hammer being tested (see 3.9.1, 3.9.2, and 3.9.3). Blows per minute shall be determined through the use of a stroboscopic light. Failure to deliver the minimum number of blows specified shall be cause for rejection.

4.4.2 Chipping test. Type II hammers shall be subjected to a test to determine compliance with table II for steel chipped. The test shall be made on a chipping hammer testing machine.

4.4.3 Service test. Upon satisfactory completion of the examination of 4.3.2, the performance tests of 4.5, the load test of 4.4.1 for type I and the chipping test of 4.4.2 for type II hammers, types I and II hammers shall satisfactorily complete 150 hours of actual work under practical service conditions. The test shall determine the ease operation, general serviceability, suitability, and durability of the tool. Failure during the service test of any components of the tool, exclusive of replaceable scaling and chipping tools shall be cause for rejection.

4.4.4 Examination after service test. Upon completion of the service test, the tool shall be disassembled and the various parts examined. Any tool having broken, cracked, or deformed parts shall be rejected. Tools satisfactorily completing the service test shall again be subjected to the performance tests of 4.5 and shall conform to the applicable requirements of this specification.

4.5 Performance tests.

4.5.1 Test conditions.

4.5.1.1 All performance tests shall be based on operation of the tools with compressed air at a gage pressure of 90-pound p.s.i. measured as close to the tool as practicable.

4.5.1.2 Air consumption shall be determined with either a liquid displacement meter or a flow-meter of common type.

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4.5.2 Operational test. Hammers shall be tested to determine correct throttle action and blows per minute delivered (see 3.9.1, 3.9.1.2, 3.9.2, 3.9.2.1, and 3.9.3, as applicable).

4.5.3 Air consumption. Hammers shall be subjected to a test for air consumption to determine compliance with 3.9.2.5, tables I and II, as applicable.

4.6 Hardness test. Each barrel and piston shall be tested for conformance with hardness requirements in accordance with method 243.1 of Fed. Std No. 151.

4.7 Inspection of preparation for delivery. Inspection of hammers prepared for delivery shall be as specified in MIL-P-12829.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Scaling and chipping hammers shall be preserved, packaged, and packed in accordance with MIL-P-12829 for each level, as specified (see 6.1).

5.2 Marking.

5.2.1 Military agencies. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.2.2 Civil agencies. Interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, style, and size (as applicable) of pneumatic hammer required (see 1.2 and table II).
(c) Diameter of shank required (see 3.10.2 (c).
(d) Accessories or extra equipment required (see 3.11).
(e) Applicable levels of preservation, packaging, and packing (see 5.1).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in QPL 00-H-116, whether or not such products have actually been so listed by that date. The attention of the supplies is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products lists is the Naval Ship Engineering Center, Department of the Navy, Washington, D.C., 20360, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Ave., Philadelphia, Pa., 19120.)

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